MATH 243 — Probability and Statistics

Joint Distributions: Joint, Marginal and Conditional Distributions, Independence, Covariance

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — joint pmf/pdf questions and how they are asked
Lesson 10 — Joint pmf and pdf, Marginal Distributions — tables, double integrals, finding k
Lesson 11 — Conditional Distributions and Independence — f(y | x), P(a < Y < b | X = x), independence tests
Lesson 12 — Expectation of Two Variables and Covariance — E(X + Y), E(XY), Cov, Var(X + Y)
Quick Notes — Joint Distributions — revision bullets

NOTE Page numbers refer to KKS Sir's Lecture 8.pdf (11 pp.), Lecture 9.pdf (pp. 5–9) and Lecture 3 & 4.pdf (Problem 5, pp. 11–12). Covariance is not on the slides but is asked in 2023-24 Q2(b); it is taught here from the textbook and marked. "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.1	Q2(b) Pens 3 red, 2 blue, 3 black, choose 2: joint pmf plot, P(X + Y ≤ 1), marginals, Var(X), Cov(X, Y) (20) [L10]
2022-23 (MATH 243) TF p. 47	Q2(a) $f(x, y) = 10xy^2$, $0 < x < y < 1$: g(x), h(y), f(y|x); P(Y > 1/2 | X = 0.25) (15) [L11]
2020-21 (MATH 245) TF p. 37	Q5(c) Tyre pressures $f = k(x^2 + y^2)$, $30 \le x, y < 50$: k; P(30 ≤ X ≤ 40, 40 ≤ Y < 50); P(both underfilled) (10) [L10]
2012-13 (MATH 241) TF p. 5	Q8(a) Joint density $(6 - x - y)/k$ on $0<x<2$, $2<y<4$: find P(1 < Y < 3 | X = 2) (18) [L11]

EXAM FOCUS 4 questions, 63 marks (45 marks since 2019-20). Every MATH 243 paper has one 15–20-mark joint-distribution question in Section A: a discrete pens/balls table (2023-24, the same as Lecture 3 & 4 Problem 5) or a continuous density with marginals and a conditional probability (2022-23, 2020-21, 2012-13). Learn the table layout, the limits of integration for triangular regions ($0 < x < y < 1$), and $\text{Cov} = E(XY) - E(X)E(Y)$.

Lesson 10 — Joint pmf and pdf, Marginal Distributions

Lec 8 pp. 1–6 · Lec 3 & 4 pp. 11–12

1. Discrete Joint Distribution (Lec 8 p. 1)

$$f(x, y) = P(X = x, Y = y)$$
$$f(x, y) \geq 0, \quad \Sigma_x \Sigma_y f(x, y) = 1, \quad P[(X, Y) \in A] = \Sigma \Sigma_{(x,y)\in A} f(x, y)$$

Example — three tosses: X = heads, Y = tails in the last two tosses (Lec 8 pp. 1–2)

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	3	2	2	1	2	1	1	0
Y	0	1	1	2	0	1	1	2

	X = 0	X = 1	X = 2	X = 3	h(y)
Y = 0	0	0	1/8	1/8	2/8
Y = 1	0	2/8	2/8	0	4/8
Y = 2	1/8	1/8	0	0	2/8
g(x)	1/8	3/8	3/8	1/8	1

2. Continuous Joint Density (Lec 8 pp. 2–3)

$$f(x, y) \geq 0, \quad \iint f(x, y)\, dx\, dy = 1, \quad P[(X, Y) \in A] = \iint_A f(x, y)\, dx\, dy$$
$$F(x, y) = P(X \leq x, Y \leq y); \quad \partial^2 F/\partial x \partial y = f(x, y)$$

$f(x, y) = x^2 + xy/3$ on $0 \leq x \leq 1$, $0 \leq y \leq 2$: $\int_0^2 (1/3 + y/6)dy = [y/3 + y^2/12]_0^2 = 1 \rightarrow$ a density. $f(x, y) = x(1 + 3y^2)/4$ on $0 \leq x \leq 2$, $0 \leq y \leq 1$: $P(0 < X < 1, 1/4 < Y < 1/2) = \int (1/8 + 3y^2/8)dy$ over $(1/4, 1/2) = 23/512$.

3. Marginal Distributions (Lec 8 pp. 4–6)

$$\text{Discrete: } g(x) = \Sigma_y f(x, y), \quad h(y) = \Sigma_x f(x, y)$$
$$\text{Continuous: } g(x) = \int f(x, y)\, dy, \quad h(y) = \int f(x, y)\, dx$$

$f(x, y) = (6 - x - y)/8$, $0 < x < 2$, $2 < y < 4$: $g(x) = -(x - 3)/4 = (3 - x)/4$, $h(y) = (5 - y)/4$; both integrate to 1. $P(X + Y < 3) = (1/8)\int_0^1 \int_2^{3-x} (6 - x - y)dy\, dx = 5/24$; $P(X < 3/2, Y < 5/2) = 9/32$.

TRAP For $P(X + Y < 3)$ the region is x from 0 to 1 only (y must exceed 2, so $x < 1$). The slide writes the outer limit as 0 to 2 but its integrand and the answer 1/24 use 0 to 1. Always sketch the region first.

4. Discrete Example — Pens (Lec 3 & 4 pp. 11–12)

Two pens from 3 blue, 2 red, 3 green; X = blue, Y = red. $f(x, y) = 3Cx \cdot 2Cy \cdot 3C(2-x-y) / 8C2$, $8C2 = 28$.

	Y = 0	Y = 1	Y = 2	g(x)
X = 0	3/28	6/28	1/28	10/28
X = 1	9/28	6/28	–	15/28
X = 2	3/28	–	–	3/28
h(y)	15/28	12/28	1/28	1

$P(X + Y \leq 1) = f(0,0) + f(1,0) + f(0,1) = (3 + 9 + 6)/28 = 9/14$

NOTE The slide's marginal row prints $P(X = 0) = P(X = 1) = P(X = 2) = 10/28 = 5/14$; the correct marginal of X is 10/28, 15/28, 3/28 (read the row totals). The marginal of Y (15/28, 12/28, 1/28) is right.

Exam Approach

1. Discrete: list every (x, y) with x + y ≤ sample size; f = product of C's ÷ total; draw the table with margins.
2. Continuous: sketch the region; find k from the double integral = 1.
3. Marginal: integrate out the other variable over its full range (which may depend on x).
4. Probability over a region: set the inner limits from the region boundary.

Practice — Lesson 10

(a) Actual — 2023-24 Q2(b) (22 batch p. 1). (b) Actual — 2020-21 Q5(c) (TF p. 37).

(a) From a bag of 3 red, 2 blue and 3 black pens a random sample of two pens is selected. X = number of red pens, Y = number of blue pens. (i) Plot f(x, y) versus (x, y). (ii) P[(X, Y) ∈ A], A = {(x, y): x + y ≤ 1}. (iii) Marginal distributions of X and Y. (iv) Variance of X. (v) Covariance between X and Y. (20)

(b) Tyre pressures X (right) and Y (left) have joint density f(x, y) = k(x² + y²), 30 ≤ x < 50, 30 ≤ y < 50 (target 40 psi). (i) Find k. (ii) Find P(30 ≤ X ≤ 40 and 40 ≤ Y < 50). (iii) Find the probability that both tyres are underfilled. (10)

✓ Solution

(a) f(x, y) = 3Cx · 2Cy · 3C(2 − x − y) / 8C2 for x, y = 0, 1, 2 with x + y ≤ 2; 8C2 = 28.

```
(i) Joint pmf (plot: a vertical stick of height f at each point (x, y))
            Y = 0    Y = 1    Y = 2    g(x)
X = 0        3/28     3/14     1/28    5/14
X = 1        9/28     3/14      –      15/28
X = 2        3/28      –        –       3/28
h(y)        15/28    3/7     1/28      1

  f  ^   9/28 |       ·(1,0)
     |   6/28 |  ·(0,1)  ·(1,1)
     |   3/28 |  ·(0,0)          ·(2,0)
     |   1/28 |         ·(0,2)
     +-----------------------------------> points (x, y)
```

```
(ii)  P(X + Y ≤ 1) = f(0,0) + f(0,1) + f(1,0) = 3/28 + 6/28 + 9/28 = 18/28 = 9/14
(iii) g(0) = 5/14, g(1) = 15/28, g(2) = 3/28;   h(0) = 15/28, h(1) = 3/7, h(2) = 1/28
(iv)  E(X) = 0 + 15/28 + 2(3/28) = 21/28 = 3/4;  E(X²) = 15/28 + 4(3/28) = 27/28
      Var(X) = 27/28 − (3/4)² = 27/28 − 9/16 = 45/112 = 0.4018
(v)   E(Y) = 12/28 + 2(1/28) = 1/2;  E(XY) = 1·1·f(1,1) = 3/14
      Cov(X, Y) = E(XY) − E(X)E(Y) = 3/14 − (3/4)(1/2) = -9/56 = −0.1607
```

The covariance is negative: with only two pens drawn, more red pens leave less room for blue ones.

(b)

```
(i)   ∫30^50 ∫30^50 (x² + y²) dx dy = 2 × 20 × ∫30^50 x² dx = 40 × (50³ − 30³)/3 = 40 × 98000/3
= 3,920,000/3
      k = 3/3,920,000 = 3/3920000  (≈ 7.65 × 10^−7)
(ii)  ∫40^50 ∫30^40 (x² + y²) dx dy = 10 ∫30^40 x² dx + 10 ∫40^50 y² dy = 10(37000/3) +
10(61000/3) = 980000/3
      P = k × 980000/3 = 980000/3920000 = 1/4 = 0.25
(iii) underfilled = below 40 psi:  ∫30^40 ∫30^40 (x² + y²) dx dy = 2 × 10 × 37000/3 = 740000/3
      P(X < 40, Y < 40) = 740000/3920000 = 37/196 = 0.1888
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Lesson 11 — Conditional Distributions and Independence

Lec 8 pp. 7–11

1. Conditional Distributions (Lec 8 p. 7)

$$f(y \mid x) = f(x, y) / g(x), \quad g(x) > 0$$
$$f(x \mid y) = f(x, y) / h(y), \quad h(y) > 0$$
$$P(a < X < b \mid Y = y) = \Sigma\, f(x \mid y) \quad \text{or} \quad \int_a^b f(x \mid y)\, dx$$

Discrete example (Lec 8 p. 8, pens table): $h(1) = 6/28 + 6/28 + 0 = 3/7$, so $f(x \mid 1) = (7/3) f(x, 1)$: $f(0 \mid 1) = 1/2$, $f(1 \mid 1) = 1/2$, $f(2 \mid 1) = 0$.

> **NOTE** In the Lec 8 p. 8 table the entry $f(1, 0)$ is printed as $3/28$; the pens table (Lesson 10 §4) has $9/28$. The conditional $f(x \mid 1)$ does not use that cell, so the slide's answer is unaffected.

Continuous example (Lec 8 pp. 8–9): $f(x, y) = (6/5)(x + y^2)$ on the unit square. $h(y) = (6/5)(1/2 + y^2)$; $f(x \mid y) = 2(x + y^2)/(1 + 2y^2)$; $f(x \mid 0.5) = (4x + 1)/3$; $P(X < 0.5 \mid Y = 0.5) = (1/3)[2x^2 + x]_0^{0.5} = 1/3$.

2. Independence (Lec 8 pp. 9–11)

$$X, Y \text{ independent} \iff f(x, y) = g(x)\, h(y) \text{ for all } (x, y) \iff f(x \mid y) = g(x)$$

- Table example (Lec 8 p. 10): every cell equals the product of its margins (e.g. $f(2, 1) = 0.10 = 0.40 \times 0.25$) → independent.
- $f(x, y) = (x + y)/8$ on $(0, 2)^2$: $g(x) = (x + 1)/4$, $h(y) = (y + 1)/4$, $g\,h = (x + 1)(y + 1)/16 \neq f$ → not independent. (The slide writes $g(x) = (x + y)/4$ by a slip; integrating out y gives $(x + 1)/4$.)

> **TRAP** For a region like $0 < x < y < 1$ the variables can never be independent: the range of Y depends on x. Write the conditional range too ($f(y \mid x)$ lives on $x < y < 1$).

Exam Approach

1. Find the marginal of the conditioning variable ($g(x)$ for $f(y \mid x)$).
2. Divide: $f(y \mid x) = f(x, y)/g(x)$; state the range of y for that x.
3. Substitute the given x, then integrate (or sum) over the requested y-interval, clipped to the range.
4. Independence: compare $f(x, y)$ with $g(x)h(y)$ at every point (one failure is enough for "not independent").

Practice — Lesson 11

(a) Actual — 2022-23 Q2(a) (TF p. 47). (b) Actual — 2012-13 Q8(a) (TF p. 5).

(a) The joint density of (X, Y), X = unit temperature change and Y = proportion of spectrum shift, is $f(x, y) = 10xy^2$, $0 < x < y < 1$. (i) Find $g(x)$, $h(y)$ and $f(y \mid x)$. (ii) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit. (15)

(b) For the joint density $f(x, y) = (6 - x - y)/c$, $0 < x < 2$, $2 < y < 4$ (printed with $c = 3$; the normalising constant is 8), find $P(1 < Y < 3 \mid X = 2)$. (18)

✓ Solution

(a) Region: $0 < x < y < 1$ (triangle above the line $y = x$).

```
(i)  g(x) = ∫_x^1 10xy² dy = (10x/3)(1 − x³),   0 < x < 1
     h(y) = ∫_0^y 10xy² dx = 5y² · y² = 5y⁴,       0 < y < 1
     f(y | x) = f(x, y)/g(x) = 10xy² / [(10x/3)(1 − x³)] = 3y²/(1 − x³),   x < y < 1
(ii) X = 0.25: f(y | 0.25) = 3y²/(1 − 1/64) = (64/21)y²,  0.25 < y < 1
     P(Y > 1/2 | X = 0.25) = ∫_1/2^1 (64/21)y² dy = (64/63)(1 − 1/8) = 8/9 = 0.8889
```

(b) The constant cancels in a conditional density, so the answer is the same for c = 3 or 8.

```
f(2, y) = (4 − y)/c;  g(2) = ∫_2^4 (4 − y)/c dy = 2/c
f(y | x = 2) = (4 − y)/2,  2 < y < 4
P(1 < Y < 3 | X = 2) = P(2 < Y < 3 | X = 2)      (Y cannot be below 2)
                     = ∫_2^3 (4 − y)/2 dy = (1/2)[4y − y²/2]_2^3 = (1/2)(7.5 − 6) = 3/4
```

NOTE x = 2 lies on the boundary of 0 < x < 2; read it as the limit x → 2 (the formula is continuous). With c = 8, ∬ f = 1 checks; with c = 3 the function is not a density — say so in one line.

Lesson 12 — Expectation of Two Variables and Covariance

Lec 9 pp. 5–9 · covariance: textbook

1. Expected Value of a Function of (X, Y) (Lec 9 pp. 5–6)

$$E[w(X, Y)] = \sum\sum w(x, y) f(x, y) \quad \text{or} \quad \iint w(x, y) f(x, y)\, dx\, dy$$
$$E(X + Y) = E(X) + E(Y); \quad E(aX + bY) = aE(X) + bE(Y)$$
$$X, Y \text{ independent} \Longrightarrow E(XY) = E(X)E(Y)$$

2. Worked Example (Lec 9 pp. 7–9)

```
f(x, y) = 2(x + y − 2xy), 0 < x, y < 1
g(x) = 2∫0^1 (x + y − 2xy) dy = 2(x + 1/2 − x) = 1   → E(X) = ∫0^1 x dx = 1/2;  likewise h(y) = 1, E(Y) = 1/2
E(X + Y) = ∫∫ (x + y) f dx dy = 1 = E(X) + E(Y) ✓
E(XY) = ∫∫ xy f dx dy = 2/9 ≠ E(X)E(Y) = 1/4  → not independent
```

Note the converse: g(x)h(y) = 1 ≠ f(x, y), so the marginals being uniform does not make X and Y independent. Example f(x, y) = 4xy on the unit square: E(X) = 4∫∫x²y = 2/3, E(Y) = 2/3.

> **NOTE** The 4xy example on the slide does not state the range of y; it is 0 < y < 1.

3. Covariance and Variance of a Sum (textbook — asked 2023-24 Q2(b))

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y)$$
$$Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab\, Cov(X, Y)$$

- Cov > 0: large X goes with large Y; Cov < 0: large X with small Y; independent ⟹ Cov = 0 (not conversely).
- Correlation coefficient $\rho = Cov(X, Y)/(\sigma_X\sigma_Y)$, $-1 \le \rho \le 1$ (the sample version is Pearson's r in the Statistics notes).

Exam Approach

1. Find the marginals, then E(X), E(Y) (and E(X²) for a variance).
2. E(XY) = Σ xy f(x, y) over cells with x, y ≠ 0 (or the double integral).
3. Cov = E(XY) − E(X)E(Y); interpret its sign.

Practice — Lesson 12

(a) Adapted — textbook Review Exercise 4.80 (other distributions p. 2). (b) Constructed.

(a) Find the covariance of X and Y with joint density f(x, y) = x + y, 0 < x < 1, 0 < y < 1.

(b) For the three-toss table of Lesson 10 (X = heads, Y = tails in the last two tosses), find Cov(X, Y).

✓ Solution

```
(a) E(X) = ∫∫ x(x + y) = 7/12, E(Y) = 7/12 (symmetry), E(XY) = ∫∫ xy(x + y) = 1/3
    Cov = 1/3 − (7/12)² = -1/144 ≈ −0.0069
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(b) E(X) = 3/2, E(Y) = 1, E(XY) = 2·1·(2/8) + 1·2·(1/8) + 1·1·(2/8) = 1
    Cov = 1 − (3/2)(1) = -1/2   (more heads → fewer tails, as expected)
```

Quick Notes — Joint Distributions

Revision points for Lessons 10–12.

L10 · Joint and Marginal

- $f(x, y) \geq 0$, $\Sigma\Sigma$ or $\iint = 1$; marginals by summing/integrating out the other variable.
- Pens 3R 2B 3K, two drawn: 3/28, 6/28, 1/28 | 9/28, 6/28 | 3/28; $P(X + Y \leq 1) = 9/14$.
- $\text{Var}(X) = 45/112$; $\text{Cov} = 3/14 - 3/8 = -9/56$.
- Tyres $k(x^2 + y^2)$ on $[30, 50)^2$: $k = 3/3{,}920{,}000$; 1/4; both under 40: 37/196.

L11 · Conditional and Independence

- $f(y \mid x) = f(x, y)/g(x)$ on the conditional range.
- $10xy^2$, $0 < x < y < 1$: $g = (10/3)x(1 - x^3)$, $h = 5y^4$, $f(y \mid x) = 3y^2/(1 - x^3)$; 8/9.
- $(6 - x - y)/8$: $f(y \mid 2) = (4 - y)/2$; $P(Y < 3 \mid X = 2) = 3/4$.
- Independent $\Leftrightarrow f = g \cdot h$ everywhere; triangular regions are never independent.

L12 · Expectation and Covariance

- $E(X + Y) = E(X) + E(Y)$ always; $E(XY) = E(X)E(Y)$ if independent.
- $\text{Cov} = E(XY) - E(X)E(Y)$; $\text{Var}(X + Y) = \text{Var}\, X + \text{Var}\, Y + 2\text{Cov}$.
- $2(x + y - 2xy)$: $E(XY) = 2/9 \neq 1/4 \rightarrow$ dependent. $x + y$: $\text{Cov} = -1/144$.